Physics and Category Theory

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Abstract

Extremely rough notes on physics and category theory. Maybe these will turn into something coherent, maybe not!

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1 Fusion Categories

1.1 Basic Definitions

Here we mostly follow [**?**]. A fusion category consists of a collection $\mathcal{C}$ (we will use the same symbol to refer to the category itself) plus a pair of binary operations

$$\oplus, \otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C} \tag{1}$$

which act roughly like the corresponding operations on familiar algebraic structures. Apparently there is a nice careful exposition of all this in [**?**]. For now, we just note that we have equalities[1]

$$x \oplus y = y \oplus x, \quad x \otimes y = y \otimes x, \quad (x \oplus y) \oplus z = x \oplus (y \oplus z). \tag{2}$$

So both operations are abelian, and the direct sum is associative. The product is not generally associative, but it is up to a certain natural isomorphism which we'll describe shortly. Finally, there exists a collection $\mathscr{I}$ of so-called simple objects, with the property that any object $x \in \mathcal{C}$ can be decomposed uniquely as a direct sum

$$x = \bigoplus_{a \in \mathscr{I}} N^a a \tag{3}$$

where necessarily[2] $N^a \in \mathbb{Z}_{\geq 0}$. Like a basis for a vector space, this collection of simple objects is generally non-unique, though apparently its cardinality is an invariant; we call the latter the *rank* of $\mathcal{C}$. After making of a choice of simple objects, the action of the direct product is charactized exactly by the *fusion coefficients* appearing in the decompositions

$$a \otimes b = \bigoplus_{c} N_c^{ab} c \tag{4}$$

ranging over all triples (a, b, c) of simple objects. Among the simple objects there exists a distiguished unit object $1 \in \mathscr{I}$ with the property that

$$a \otimes 1 = 1 \otimes a = 1 \ \forall a \in \mathscr{I}. \tag{5}$$

Morever, every $a \in \mathscr{I}$ has a unique dual $\bar{a} \in \mathscr{I}$ obeying

$$N_{a\bar{a}}^1 = N_{\bar{a}a}^1 = 1. \tag{6}$$

[1] As we'll see in our favorite example, for this to be literal equality we may need the objects in $\mathcal{C}$ to be isomorphism classes of our favorite objects.

[2] A colleague tells me that this subfield feels strongly that $0 \notin \mathbb{N}$.

1.2 Example: Representations of a Fixed Group

The simplest nontrivial example of all of this arises by fixing a group G and letting $\mathcal{C}$ consist of the representations[3] $\rho : G \to \mathrm{End}(V)$ for V a $\mathbb{C}$-linear space. Then $\oplus, \otimes$ are the usual sum and product of representations, the natural simple objects are the irreducible representations, and the fusion coefficients are Clebsch-Gordon coefficients. The unit simple object is the trivial constant scalar map

$$g \in G \mapsto 1 \in \mathbb{C}^* \simeq \mathrm{End}(\mathbb{C}) \tag{7}$$

The nature of dual objects requires slightly more work. Given any irrep $\rho : G \to \mathrm{End}(V)$, or in the present notation $a = (V, \rho)$, the dual object is $\bar{a} = (V^*, \rho^\vee)$, where V^* is the usual dual space

$$V^* = \{\text{linear maps } V \to \mathbb{C}\} = \mathrm{End}(V, \mathbb{C}) \tag{8}$$

and $\rho^\vee(g)$ is the inverse transpose, or equivalently acts on dual vectors by pulling back under the inverse, or equivalently

$$\rho^\vee(g)(\lambda) = \lambda \circ \rho(g^{-1}) \in V^* \text{ for } \lambda \in V^*. \tag{9}$$

In matrix notation,

$$\rho^\vee(g) = (\rho(g^{-1}))^T. \tag{10}$$

Now consider the product of these simple objects

$$a \otimes \bar{a} = (V \otimes V^*, \rho \otimes \rho^\vee). \tag{11}$$

The canonical evaluation map

$$E : V \otimes V^* \to \mathbb{C}, \quad (v, \lambda) \mapsto \lambda(v) \tag{12}$$

is G-invariant in this representation, in the sense that[4]

$$v \otimes \lambda \xmapsto{(\rho \otimes \rho^\vee)(g)} \rho(g)(v) \otimes \rho^T(g^{-1})(\lambda) \xmapsto{E} \left(\lambda \circ \rho(g^{-1})\right)(\rho(g)v) = \lambda(v). \tag{13}$$

More briefly,

$$E \circ (\rho \otimes \rho^\vee)(g) = E. \tag{14}$$

As already noted, finding the fusion coefficients here amounts to finding the Clebsch-Gordon coefficients.

[3] Modulo conjugation equivalence, if we are being careful about equality vs isomorphism

[4] Here we calculate only on product pairs, but by linearity this extends to general vectors in the product representation